

Universal Mixing Rule for Cubic Equations of State Applicable to Symmetric and Asymmetric Systems: Results with the Peng–Robinson Equation of State

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A mixing rule for cubic equations of state (CEoS) applicable to all types of system asymmetries—referred to hereafter as the universal mixing rule (UMR)—is proposed. For the cohesion parameter of the CEoS, the mixing rule involves the Staverman–Guggenheim part of the combinatorial term and the residual term of the original UNIFAC Gibbs free energy expression. For the covolume parameter of the CEoS, the quadratic concentration-dependent mixing rule is used with the combining rule for the cross parameter $b_{ij} = [^{1/2}(b_i^{1/2} + b_j^{1/2})]^2$. This UMR is applied to the volume-translated and modified version of the Peng–Robinson equation of state of Magoulas and Tassios (*Fluid Phase Equilib.* **1990**, 56, 119), leading to what is referred to as the UMR–PR model. Very satisfactory results are obtained using the existing interaction parameters of the original UNIFAC model for vapor–liquid equilibrium predictions at low and high pressures for a wide range of system asymmetries including mixtures containing polymers. Satisfactory liquid–liquid equilibrium predictions are also obtained with the UMR–PR model.

1. Introduction

Accurate description of phase equilibria is a key step for the successful process design. Utilization of cubic equations of state (CEoS) is a widespread approach for this purpose, as they have the advantage of being applicable over wide ranges of temperature and pressure and for mixtures of various components, from light gases to heavy liquids. Because the same equation of state is applied to the various phases at equilibrium, a consistent description is obtained, and questions of standard states do not arise as in the case of the other popular approach, the so-called γ – φ approach.

Cubic equations of state coupled with the conventional van der Waals one-fluid (vdW1f) mixing rules are, however, limited to nonpolar fluids. The vdW1f mixing rules are incapable of representing the highly nonideal mixture behavior of polar or associating fluids. A number of mixing rules have been proposed to extend the applicability of equations of state to highly nonideal mixtures with varying degrees of success.¹ In the past two decades, mixing rules for cubic equations of state derived from excess Gibbs free energy expressions have been a subject of special interest,^{1–5} and several such models, which are called EoS/ G^E models, have been proposed. Of special interest are those models that couple a cubic EoS with UNIFAC because the resulting models are purely predictive tools.^{6–9}

Very recently, Gmehling and co-workers¹⁰ proposed an EoS/ G^E model, the so-called GCEOS–VTPR model, that combines a volume-translated PR EoS with the original UNIFAC model.¹¹ Using an empirical consideration through the so-called “effective” van der Waals

volume parameters for alkane groups, which were introduced by Li et al.¹² in the PSRK model, they eliminated the two Flory–Huggins-type combinatorial contributions originating from the EoS and the original UNIFAC model. They also eliminated the Staverman–Guggenheim part of the original UNIFAC combinatorial term on grounds of its negligible contribution. Finally, they empirically introduced an exponent of 0.75 in the combining rule for the cross covolume parameter of the EoS, b_{ij} , involved in the quadratic mixing rule for the mixture covolume parameter, b . They proceeded then to the reevaluation of quadratic temperature-dependent UNIFAC interaction parameters for gas/alkane mixtures and mixtures containing benzene and ketones.^{10,13} Later, they applied their model to low-pressure solvent/polymer vapor–liquid equilibrium (VLE)¹⁴ using a different exponent in the b_{ij} parameter, equal to 0.5, and the interaction parameters of the original UNIFAC model.

In this work, we propose a universal mixing rule (UMR) for cubic equations of state applicable to both symmetric and asymmetric systems. Results with the UMR are presented for the EoS of Tassios and Magoulas (t-mPR EoS)¹⁵ (UMR–PR model). It is demonstrated that the UMR–PR model can utilize the existing interaction parameter table of the original UNIFAC model, yielding very satisfactory phase equilibrium predictions for both symmetric and asymmetric systems including systems containing polymers.

2. Development of the Universal Mixing Rule (UMR) and the UMR–PR Model

Michelsen^{4,5} proposed the following mixing rules, the so-called MHV1 mixing rules, for the mixture attractive and repulsive term parameters of a cubic EoS, based

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on the zero-pressure reference state approach

$$\alpha = \frac{1}{A} \left(\frac{G_{AC}^E}{RT} + \sum_i x_i \ln \frac{b}{b_i} \right) + \sum_i x_i \alpha_i \quad (1)$$

$$b = \sum_i \sum_j x_i x_j b_{ij} \quad \text{and} \quad b_{ij} = \left(\frac{b_i^{1/s} + b_j^{1/s}}{2} \right)^s \quad (2)$$

The dimensionless quantity a in eq 1 is the cubic EoS cohesion parameter, which is defined as $\alpha = a/bRT$, where a and b are the mixture attractive and covolume parameters of the EoS, respectively; R is the gas constant; and T is the temperature. Similarly, for the pure component i , $\alpha_i = a_i/b_iRT$. For the mixture parameter b , Michelsen proposed the use of $s = 1$ in eq 2, which results to the classical arithmetic mean mixing rule. Also, A in eq 1 is a constant that depends on the EoS used—it has the value of 0.53 for the PR EoS—and G_{AC}^E is the value of G^E obtained from an activity coefficient model such as the original UNIFAC.

The mixing rule of eq 1 has been demonstrated to fail to satisfactorily predict phase equilibria in athermal asymmetric systems,⁹ such as systems containing short-chain and long-chain alkanes, where G_{AC}^E is equal to the combinatorial term that is given by the following expression

$$G_{AC}^{E, \text{comb}} = G_{AC}^{E, \text{FH}} + G_{AC}^{E, \text{SG}} \quad (3)$$

$G_{AC}^{E, \text{FH}}$ is the Flory–Huggins (FH) contribution

$$G_{AC}^{E, \text{FH}} = \sum_i x_i \ln \frac{\phi_i}{x_i} \quad (4)$$

and $G_{AC}^{E, \text{SG}}$ is the Staverman–Guggenheim (SG) contribution

$$G_{AC}^{E, \text{SG}} = 5 \sum_i x_i q_i \ln \frac{\theta_i}{\phi_i} \quad (5)$$

$$\phi_i = \frac{r_i^p}{\sum_j x_j r_j^p} \quad (6)$$

$$\theta_i = \frac{q_i}{\sum_j x_j q_j} \quad (7)$$

The exponent p in eq 6 is equal to 1 for the original UNIFAC, while r and q are the van der Waals volume and area parameters of the molecule i , respectively, both calculated through group contribution increments given by Bondi.¹⁶

The inadequacy of the MHV1 mixing rule in asymmetric systems could be attributed to the poor performance of the original UNIFAC combinatorial term for such systems,¹⁷ as demonstrated in Figure 1 for the system *n*-hexane/*n*-hexadecane using the original UNIFAC model alone. Note that, in such systems, the SG term contribution is very small. In the same figure, the results obtained with the MHV1 mixing rule using the

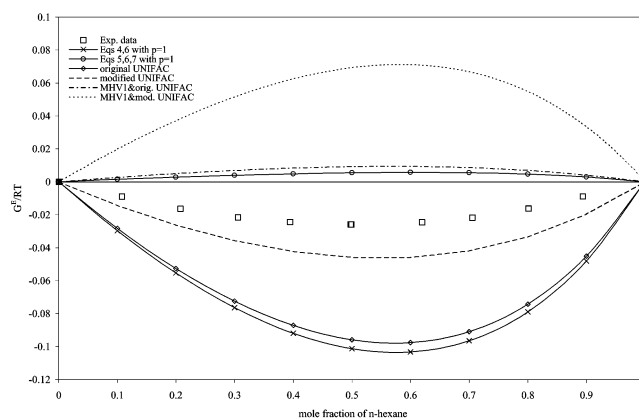


Figure 1. Experimental and predicted G^E/RT for the system *n*-hexane/*n*-hexadecane. Experimental data from Weiguo et al.³¹ The poor results of the original UNIFAC (original UNIFAC in the legend) as compared to the modified UNIFAC of Larsen et al.¹⁸ (modified UNIFAC in the legend) indicate the failure of the FH term for asymmetric systems. The results of the MHV1 mixing rule with the original UNIFAC (MHV1&orig. UNIFAC in the legend) and the modified UNIFAC one (MHV1&mod. UNIFAC in the legend) indicate the failure of the EoS-derived FH term for same systems.

t-mPR EoS and the original UNIFAC term are also included and indicate the aforementioned poor performance of the resulting model. Thus, one solution to the problem could be the use of a reliable UNIFAC combinatorial term for such systems. Such a combinatorial term is the modified UNIFAC(2/3) combinatorial term of Larsen et al.¹⁸ where the SG contribution in eq 3 has been eliminated and the exponent p in eq 6 has the value of $2/3$. The results of Figure 1 indicate the good performance of the modified UNIFAC model. However, as shown in Figure 1, combination of the MHV1 mixing rule with the t-mPR EoS and the modified UNIFAC term does not lead to better results. This indicates that the EoS-derived Flory–Huggins-type G^E combinatorial ($= \sum x_i \ln b/b_i$ that will hereafter be called the FH-EoS term) is responsible for the poor results. In other words, use of the FH part in the combinatorial term of the original UNIFAC (eqs 4 and 6 with $p = 1$) in eq 1 combines a very poor activity coefficient model combinatorial term with an equally poor EoS-derived combinatorial term. This issue has been also addressed in the work of Kontogeorgis and Vlamos.¹⁹

Actually, the LCVM model,⁹ which has proven to be the most successful EoS/ G^E model for asymmetric systems, empirically corrects this inadequacy through the incorporation of an empirical constant parameter, λ , in the mixing rule of eq 1. In the same fashion, Li et al.¹² using empirically determined van der Waals volume and area parameters for the alkane groups (CH_3 , CH_2 , CH , C) improved the performance of the PSRK model, which uses an MHV1-type mixing rule, in the prediction of VLE in asymmetric mixtures containing alkanes.

The contribution of the FH-EoS term can be modified, however, by changing the value of s in eq 2. Because of the observation of the poor performance resulting from the simultaneous use of the UNIFAC–FH combinatorial contribution and the FH-EoS one, we decided to search for a value of s that tends to cancel out these two terms. Figure 2 presents UNIFAC–FH (eq 4 with $p = 1$) and FH-EoS results for four athermal alkane systems that span a wide range of asymmetries and three s values: $4/3$, 2, and 3. Covolume parameters were calculated with

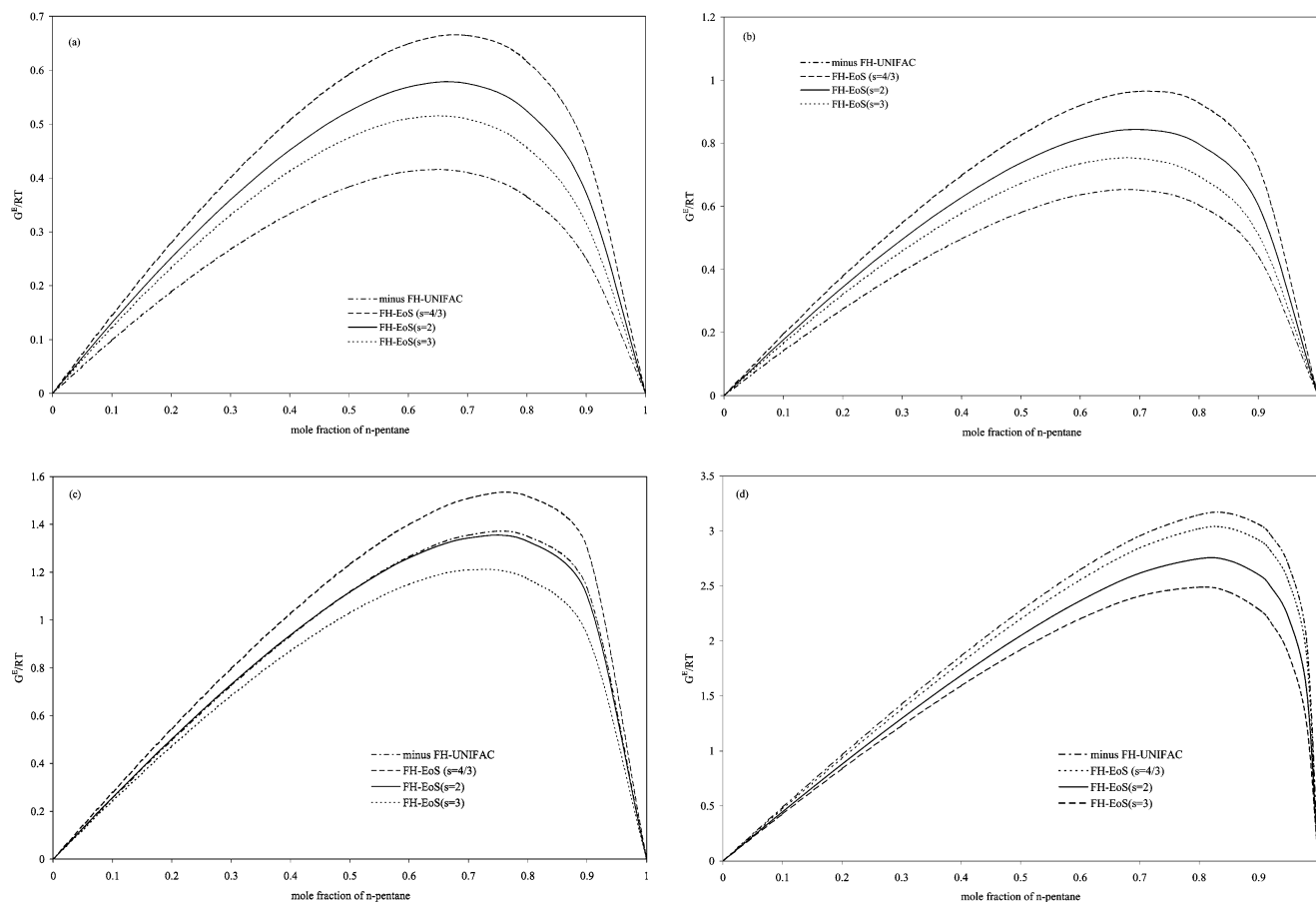


Figure 2. UNIFAC model and the EoS-derived FH contribution to G^E/RT for the systems (a) *n*-pentane/*n*-hexatriacontane, (b) *n*-pentane/*n*-C60, (c) *n*-pentane/*n*-C200, and (d) *n*-pentane/poly(ethylene) (MW = 30000).

Table 1. Flory–Huggins Contribution, Staverman–Guggenheim Contribution, and Total Combinatorial Infinite-Dilution Activity Coefficients for Some Selected Binary Systems

solute (1)	solvent (2)	r_1^a	q_1^a	r_2^a	q_2^a	$(\phi_1/\theta_1)^\infty$	$\gamma_1^{\text{FH},\infty}$	$\gamma_1^{\text{SG},\infty}$	$\gamma_1^{\text{comb},\infty b}$
methanol	benzene	1.431	1.432	3.188	2.400	0.752	0.779	1.30	1.01
CHCl ₃	acetone	2.87	2.41	2.574	2.296	1.062	0.994	1.02	1.02
methyl acetate	benzene	2.804	2.576	3.188	2.400	0.820	0.992	1.27	1.26
<i>n</i> -heptane	<i>n</i> -C20	5.174	4.396	13.941	11.416	0.964	0.696	1.01	0.71
<i>n</i> -heptane	<i>n</i> -C36	5.174	4.396	24.732	20.056	0.954	0.461	1.02	0.47
ethane	<i>n</i> -C20	1.802	1.696	13.941	11.416	0.870	0.309	1.08	0.33
ethane	<i>n</i> -C30	1.802	1.696	20.685	16.816	0.864	0.217	1.09	0.24
ethane	<i>n</i> -C44	1.802	1.696	30.127	24.376	0.860	0.153	1.10	0.17
methanol	water	1.431	1.432	0.92	1.400	1.521	0.893	2.07	1.85
ethanol	water	2.575	2.588	0.92	1.400	1.514	0.463	3.62	1.68
2-propanol	water	3.249	3.124	0.92	1.400	1.583	0.281	6.89	1.94
<i>n</i> -decanol	water	7.971	6.908	0.92	1.400	1.756	0.004	782.5	3.18
<i>n</i> -eicosanol	water	14.715	12.308	0.92	1.400	1.819	4.9E–6	7.99×10^5	3.93
<i>n</i> -decane	water	7.197	6.016	0.92	1.400	1.821	0.009	781	6.65
<i>n</i> -hexadecane	water	11.244	9.256	0.92	1.400	1.849	1.6E–4	50832	8.31
phenol	water	3.552	2.68	0.92	1.400	2.017	0.221	68.3	15.10

^a Fitted r and q values are used in original UNIFAC for water and alcohols except methanol. ^b $\gamma_2^{\text{comb},\infty}$ values for these systems are not very large because either $(\phi_2/\theta_2)^\infty$ does not deviate significantly from unity or the q_2 value is not very large.

the t-mPR EoS using experimental values for the critical properties and acentric factor for *n*-pentane²⁰ and calculated values for *n*-C36 from Magoulas and Tassios,¹⁵ for *n*-C60 and *n*-C200 from Constantinou and Gani,²¹ and for the polymer PE(30000) from Louli and Tassios.²² Unfortunately, there is no single s value that works best in the whole range of asymmetries, with $s = 2$ and $s = 3$ being the most promising values. Taking into account the latter result and some preliminary VLE calculations performed in asymmetric alkane systems, it is concluded that the value of $s = 2$ represents the best compromise and will be considered next.

Another important issue is the contribution of the SG term to the combinatorial activity coefficient predicted from the original UNIFAC model. Although the combinatorial activity coefficient should be less than unity, very large positive combinatorial activity coefficients have been observed with the original UNIFAC model.^{23,24} Because the Flory–Huggins part is always less than unity, this behavior is attributed to the Staverman–Guggenheim part, and it becomes very pronounced when fitted geometry parameters (group area and volume) are used to obtain successful VLE results for water and alcohols other than methanol, for example.

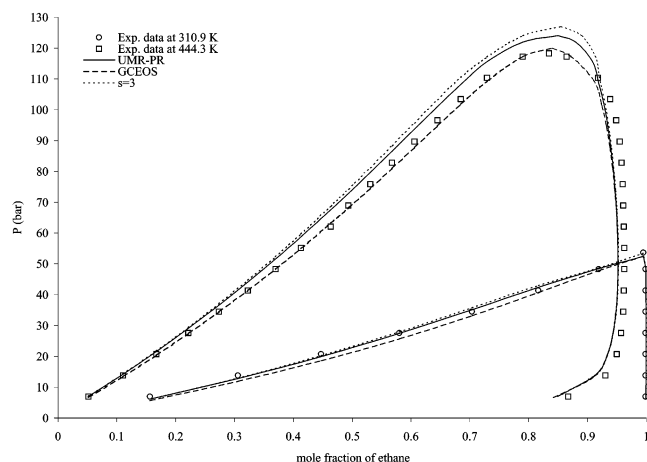


Figure 3. VLE results for the system ethane/*n*-decane. Experimental data from Reamer et al.³² Because of the relatively low asymmetry of the system, similar results are obtained by all three models.

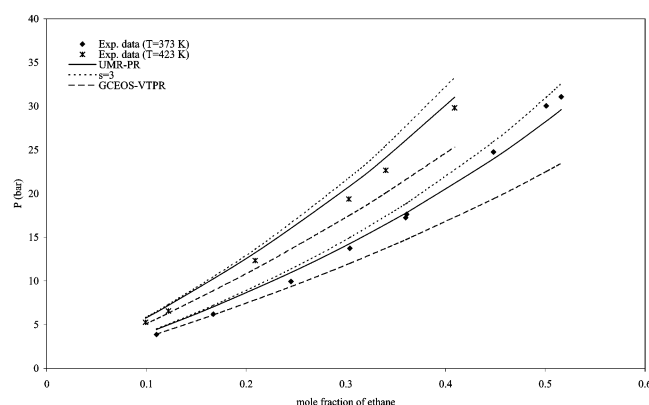


Figure 4. VLE results for the system ethane/*n*-tetratetracontane. Experimental data from Gasem et al.³³

It is clear from eq 5 that the problem is particularly pronounced when q_i is large and the ratio φ_i/θ_i deviates significantly from unity. This is shown in Table 1 with the infinite-dilution combinatorial activity coefficients for some selected binary mixtures.

On the basis of this observation, we decided to keep the SG contribution term in the G_{AC}^E expression. Noting that the FH-EoS term is the same for all CEoS that involve the $RT/(V-b)$ repulsive term, we arrive at a mixing rule applicable for all of them and for all asymmetries. We refer to it as the universal mixing rule (UMR)

$$a = \frac{1}{A} \frac{G_{AC}^{E,SG} + G_{AC}^{E,res}}{RT} + \sum_i x_i a_i \quad (8)$$

$$b = \sum_i \sum_j x_i x_j b_{ij} \quad \text{and} \quad b_{ij} = \left(\frac{b_i^{1/s} + b_j^{1/s}}{2} \right)^s \quad (9)$$

where A depends on the CEoS used, $G_{AC}^{E,res}$ is the residual part of the UNIFAC Gibbs free energy expression, and $s = 2$.

Use of this UMR with the t-mPR EoS,¹⁵ which is briefly presented in the Appendix, leads to what we will refer to hereafter as the UMR-PR model. For the t-mPR EoS, $A = 0.53$ in eq 8. We examine next the performance

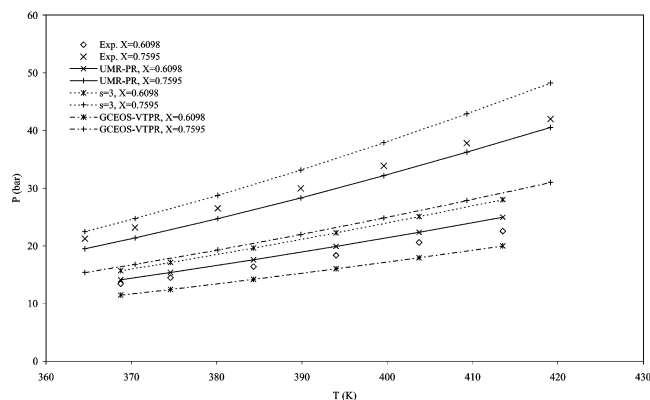


Figure 5. VLE results for the system propane/*n*-C60. Experimental data from Peters et al.³⁸

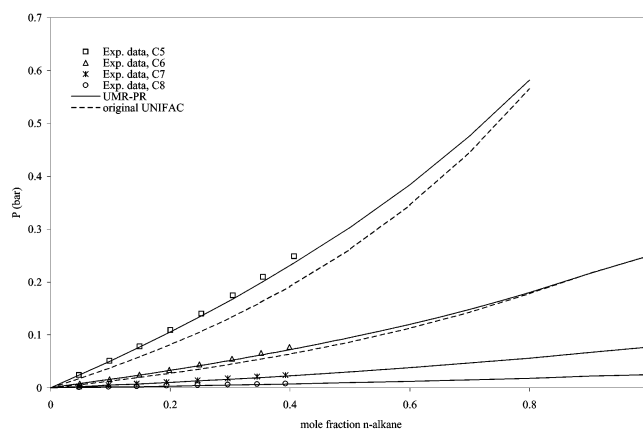


Figure 6. VLE results for *n*-alkane/squalane systems. Experimental data from Ashworth.³⁹

Table 2. VLE Results for Propane/*n*-Alkane Systems

<i>n</i> -alkane	ref	<i>T</i> range (K)	<i>P</i> range (bar)	UMR-PR ΔP (%) ^a
<i>n</i> -C8	34	334–543	6.9–59	5.2
<i>n</i> -C10	35	277–511	1.7–71	6.4
<i>n</i> -C20	36	279–358	5.5–32.5	2.6
<i>n</i> -C34	37	320–428	15.8–95.2	7.3
<i>n</i> -C60	38	357–431	13.4–57.9	11.7

^a ΔP (%) = $(1/NP) \sum_{i=1}^{NP} |(P_i^{calc} - P_i^{exp})/P_i^{exp}| \times 100$, where NP is the number of data points.

of the UMR-PR model using temperature-independent UNIFAC interaction parameters taken from Hansen et al.¹¹

3. Results and Discussion

3.1. VLE Predictions for Athermal Alkane/Alkane Systems. Systems containing only alkanes can be used as a direct test of the performance of the new EoS/ G^E model because no interaction parameters are used. Figures 3 and 4 present typical VLE predictions for ethane/alkane systems, and the same is the case in Figure 5 for the C3/C60 system, where, for comparison purposes, results with the mixing rules of eqs 8 and 9 with $s = 3$ and with the GCEOS-VTPR model of Gmehling and co-workers have been included. For all models, the same EoS (t-mPR EoS) has been used. Note that application of the GCEOS-VTPR model with the VPTR EoS, as proposed by Gmehling and co-workers, leads to practically identical results. Binary VLE predictions for propane with a series of *n*-alkanes are presented in Table 2, whereas Figure 6 presents results for *n*-alkane/squalane systems.

Table 3. VLE Results with the UMR-PR Model for Polar Systems at Low Pressures

system	ref	T (K)	P range (bar)	NP	ΔP (%) ^a	$\Delta y \times 100$ ^b
acetone/CHCl ₃	40	298.0	0.26–0.3	31	0.8	0.9
	41	308.0	0.37–0.44	31	0.6	0.4
benzene/methyl acetate	40	323.0	0.68–0.79	31	0.8	1.3
	42	303.2	0.18–0.35	16	1.6	0.3
	42	313.2	0.28–0.53	9	2.4	0.4
ethanol/ <i>n</i> -butane	43	323.2	0.39–0.78	17	1.3	0.2
	44	298.5	0.08–2.54	24	3.5	
methanol/water	44	323.5	0.30–5.04	24	4.3	
	45	298.2	0.06–0.15	10	1.9	0.5
ethanol/water	45	298.2	0.04–0.08	20	4.1	2.7
	45	348.2	0.62–0.88	9	2.3	1.9

^a $\Delta P\% = (1/NP) \sum_{i=1}^{NP} |(P_i^{\text{calc}} - P_i^{\text{exp}})/P_i^{\text{exp}}| \times 100$. ^b $\Delta y \times 100 = (100/NP) \sum_{i=1}^{NP} |y_i^{\text{exp}} - y_i^{\text{calc}}|$, where NP is the number of data points.

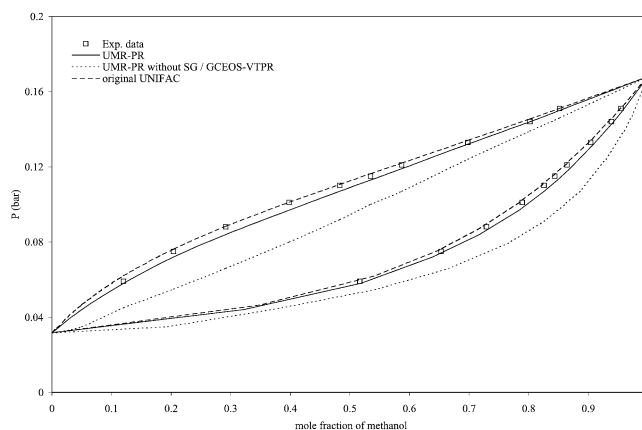
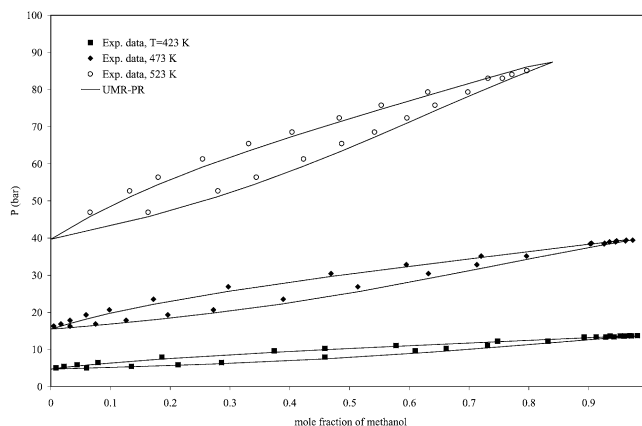
Table 4. VLE Results with the UMR-PR Model for Polar Systems at High Pressures

system	ref	T (K)	P range (bar)	NP	ΔP (%) ^a	$\Delta y \times 100$
acetone/methanol	46	373.15	3.5–4.0	14	4.3	1.2
	46	423.15	11.6–14.3	15	2.5	1.7
	46	473.15	29.5–39.8	10	5.0	3.2
methanol/benzene	47	373.15	3.1–4.2	10	6.9	2.4
	47	413.15	6.7–11.8	10	10.5	3.2
	47	493.15	22.6–57.6	10	6.7	5.7
methanol/water	46	373.15	1.0–3.4	16	2.7	1.1
	46	423.15	5.1–13.7	14	2.9	1.8
	46	473.15	16.3–39.4	15	2.6	1.0
	46	523.15	46.9–85.1	12	2.0	0.8
ethanol/water	48	423.15	5.6–9.9	17	3.6	0.6
	48	473.15	17.9–29.5	17	4.2	1.6
	48	523.15	40.8–71.7	15	5.0	1.0
	48	548.15	61.4–98.6	11	4.1	1.5
	48	573.15	88.5–125.5	7	2.5	1.0
	48	598.15	124.0–157.0	6	2.5	0.9
2-propanol/water	48	623.15	170.6–189.7	3	0.5	0.1
	48	423.15	5.2–8.6	19	3.3	1.3
	48	473.15	18.5–28.1	18	4.0	1.5
	48	523.15	43.1–66.2	13	4.0	1.5
	48	548.15	69.0–93.1	17	3.3	1.9
acetone/water	48	573.15	88.9–123.5	5	1.3	2.5
	46	423.15	5.0–12.0	17	3.4	1.7
	46	473.15	16.0–30.3	25	4.0	1.2
ethanol/ <i>n</i> -butane	46	523.15	40.4–67.6	13	1.7	0.8
	44	343.65	0.8–8.8	24	4.6	

^a $\Delta P (\%) = (1/NP) \sum_{i=1}^{NP} |(P_i^{\text{calc}} - P_i^{\text{exp}})/P_i^{\text{exp}}| \times 100$, where NP is the number of data points.

In all calculations, the attractive and covolume term parameters of the t-mPR EoS for pure fluids were determined from the critical properties and the acentric factor. Experimental critical properties and acentric factors were used for alkanes up to *n*-eicosane,²⁰ whereas for the other alkanes for which such data are not experimentally available, predicted values were employed. In accordance with the conclusions of Kontogeorgis and Tassios,²⁵ the Magoulas–Tassios method¹⁵ was used for *n*-alkanes with carbon number up to 44 and that of Constantinou–Gani²¹ for the higher carbon chain *n*-alkanes and for squalane.

Figures 3–5 indicate that the three EoS/G^E models perform satisfactorily for relatively low asymmetries (Figure 3), whereas only the UMR-PR model provides satisfactory results at the higher asymmetries. Notice, finally, that, as shown in Figure 6, the proposed mixing rule corrects the underprediction of activity coefficients of the original UNIFAC that is often used for low-pressure VLE predictions in asymmetric systems.

**Figure 7.** VLE results for the system water/methanol at 298 K. Experimental data from Gmehling et al.⁴⁶ Notice the poor results obtained in the absence of the SG term.**Figure 8.** VLE results for the system water/methanol at high pressures. Experimental data from Griswold and Wong.⁴⁶

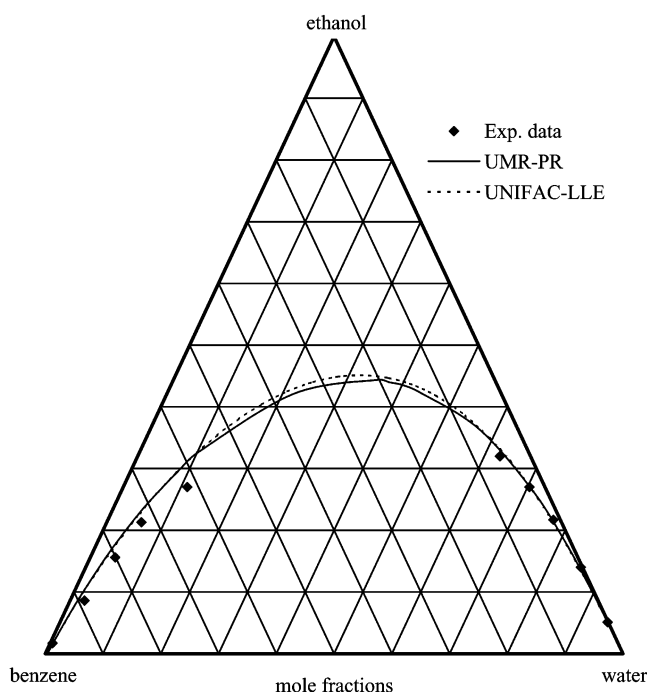
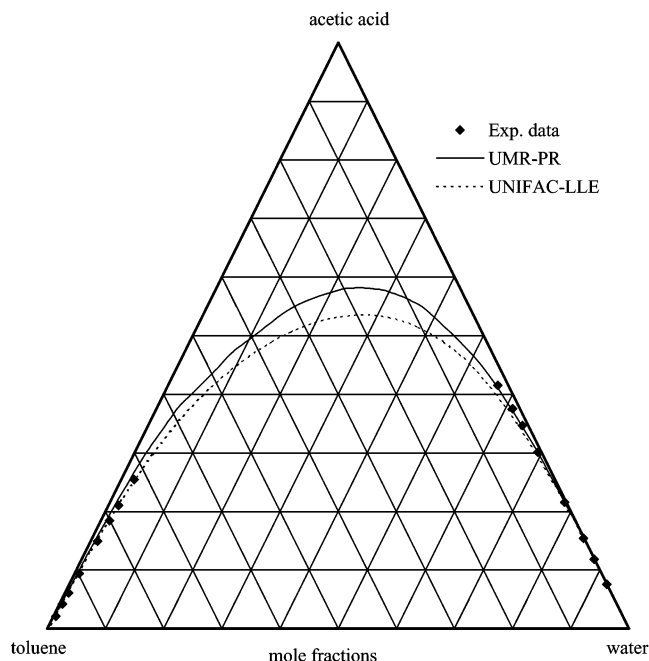
3.2. VLE Predictions for Polar Systems. Very satisfactory VLE predictions are obtained with the UMR-PR model for polar systems utilizing the existing table of interaction parameters of the original UNIFAC model.¹¹ This is shown in Tables 3 and 4 for binary systems at low and high pressures, respectively, and in Table 5 for ternary systems, and typical results are presented graphically in Figures 7 and 8. For pure polar compounds, the attractive and covolume term parameters of the t-mPR EoS were calculated using the critical properties and the Mathias–Copeman²⁶ expression for the temperature dependency of the attractive term parameter.

Figure 7 indicates that, for nearly symmetric systems, the UMR-PR model gives predictions similar to those of the original UNIFAC model. This is due to the fact that, for such systems, the combining rule for b_{ij} has no significant effect on the obtained results. Also, Figure 7 demonstrates the importance of the SG term when fitted r and q values are involved in the original UNIFAC model: removing the SG term in eq 8 leads to very poor results (UMR-PR without SG), which are identical to the results obtained with the GCEOS-VTPR model, where this term has been removed.

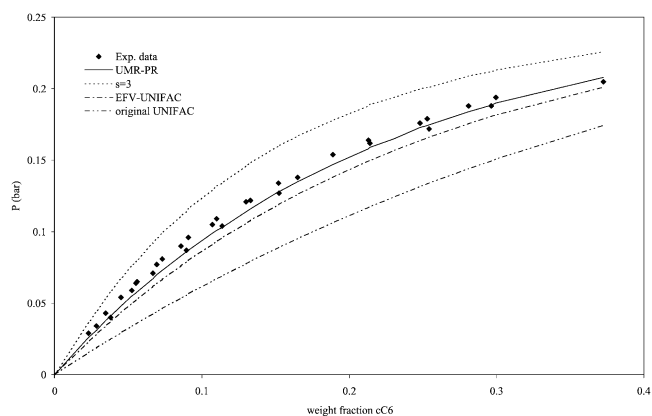
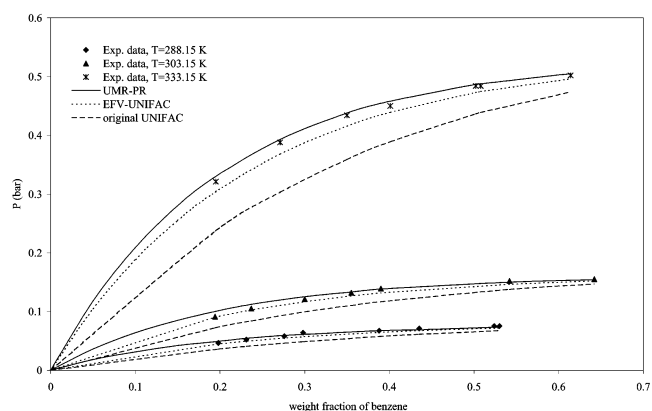
The high-pressure results of Table 3 and Figure 8, finally, are quite gratifying because they were obtained with temperature-independent interaction parameters even though temperatures over 500 K are reached. It is worth noting that the use of temperature-dependent UNIFAC interaction parameters⁵⁵ does not offer any advantage as their application here gave similar results.

Table 5. VLE Results with the UMR-PR Model for Ternary Polar Systems

system	ref	T (K)	P range (bar)	NP	ΔP (%)	$\Delta y_1 \times 100$	$\Delta y_2 \times 100$
acetone (1)/methanol (2)/water(3)	46	373.2	1.2–3.9	51	5.2	2.9	2.7
pentane (1)/methanol (2)/acetone(3)	49	372.7	5.4–8.2	36	2.2	1.9	1.7
acetone (1)/CHCl ₃ (2)/methanol(3)	50	323.2	0.6–0.9	150	1.8	0.7	0.7
benzene (1)/cyclohexane (2)/2-propanol(3)	51	~344	1.013	70	3.4	0.8	0.9

**Figure 9.** LLE results for the system benzene/water/ethanol at 318 K. Experimental data from Sorensen and Arlt.⁵²**Figure 10.** LLE results for the system acetic acid/water/toluene at 298 K. Experimental data from Sorensen and Arlt.⁵²

3.3. LLE Predictions in Polar Systems. Figures 9 and 10 present some liquid–liquid equilibrium (LLE) predictions with the UMR-PR model. Because it is well-known that UNIFAC with parameters determined from VLE data does not give satisfactory LLE predictions, the UMR-PR model has been coupled here with the

**Figure 11.** VLE results for the system poly(isobutylene) (MW = 50000)/cyclohexane at 313.2 K. Experimental data from Wohlfarth.⁵³**Figure 12.** VLE results for the system poly(styrene) (MW = 63000)/benzene. Experimental data from Hao et al.⁵⁴

UNIFAC-LLE model²⁷ that was developed especially for LLE predictions. Because the systems are nearly symmetric the UMR-PR model provides satisfactory results that are again similar to those of the UNIFAC-LLE model.

3.4. VLE Predictions in Polymer–Solvent Systems. Figures 11–15 present VLE predictions for polymer–solvent systems. The attractive and covolume term parameter of the t-mPR EoS for the solvents were calculated with the procedure described above, whereas for the polymers, they were calculated with the method of Louli and Tassios.²² An exception was the A4 dendrimer for which such parameters are not available from Louli and Tassios, so they were calculated using the density estimates given in the paper where the VLE data are presented.²⁸ In Figure 11, the results with the mixing rules of eqs 8 and 9 with $s = 2$ (UMR-PR model) and $s = 3$ indicate that the choice of $s = 2$ is the correct one for very asymmetric systems as well. Furthermore, for comparison purposes, Figures 11–13 include prediction results obtained with the original UNIFAC and the EFV-UNIFAC models. The EFV-UNIFAC model couples the combinatorial/free volume term proposed by Elbro et al.²⁹ with the original UNIFAC residual term,

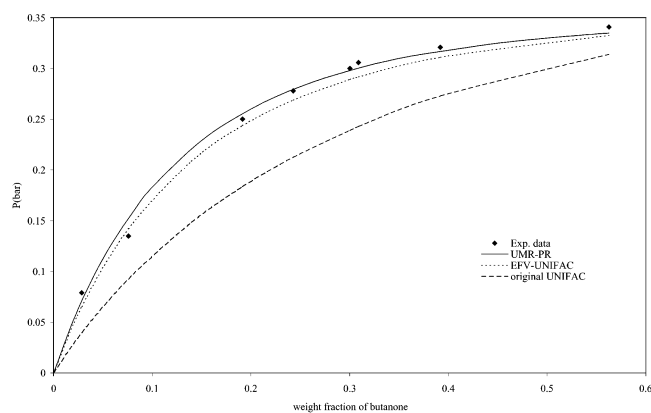


Figure 13. VLE results for the system poly(methyl methacrylate) (MW = 19770)/butanone at 322 K. Experimental data from Hao et al.⁵⁴

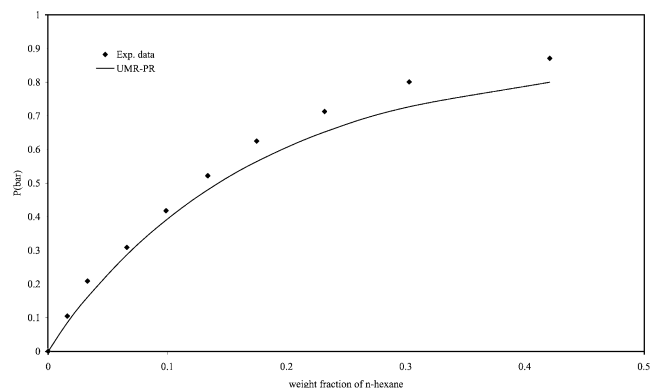


Figure 14. VLE results for the dendrimer A4 (MW = 8870)/n-hexane at 338 K. Experimental data from Lieu et al.²⁸

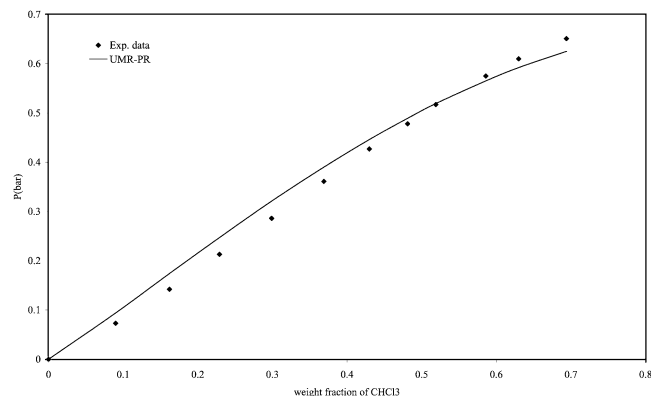


Figure 15. VLE results for the dendrimer A4 (MW = 8870)/chloroform at 323 K. Experimental data from Lieu et al.²⁸

and it has been successfully applied to VLE predictions of low-pressure polymer–solvent systems.³⁰ The UMR–PR model gives very satisfactory predictions, similar to those of the EFV–UNIFAC model. Notice that, as with the case of the *n*-alkane/squalane systems (Figure 6), the UMR–PR model again corrects the underprediction of the original UNIFAC model. Finally, the results of Figures 14 and 15 for the A4 dendrimer are quite satisfactory considering the type of polymer and the fact that density estimates for this polymer were used.

4. Conclusions

A mixing rule for the attractive and repulsive terms of cubic equations of state is proposed in this study. The Staverman–Guggenheim term of the combinatorial part

and the residual part of the original UNIFAC Gibbs free energy expression are involved in the mixing rule for the cubic EoS cohesion parameter, whereas the quadratic concentration-dependent mixing rule is used for the covolume parameter of the EoS with a modified combining rule for the cross parameter. It should be noted that any cubic EoS with $RT/(V - b)$ as the repulsive term can be used with the new mixing rules, whereas the Gibbs free energy expression used must be only the original UNIFAC or, of course, the original UNIQUAC⁵⁶ when the model is going to be used as a correlative tool. Furthermore, this mixing rule is applicable to both symmetric and asymmetric systems including polymers. It is referred to, therefore, as the universal mixing rule (UMR) for cubic equations of state.

This mixing rule is used with the t-mPR EoS leading to a new EoS/ G^E model, referred to as UMR–PR. Application of this model over a wide range of temperatures, pressures, and system asymmetries yielded very satisfactory phase equilibrium predictions. Furthermore, the results indicate that the new model is purely predictive as it is able to utilize successfully the existing original UNIFAC model interaction parameters and, consequently, there is no need for reevaluation of them.

Appendix: t-mPR EoS

The t-mPR EoS¹⁵ in terms of pressure is given by the following expression

$$P = \frac{RT}{V + t - b} - \frac{a}{(V + t)(V + t + b) + b(V + t - b)} \quad (\text{A1})$$

The pure-component parameter a is given by

$$a = 0.45724 \frac{R^2 T_c^2}{P_c} f(T_r) \quad (\text{A2})$$

where T_c is the critical temperature, P_c is the critical pressure, and T_r is the reduced temperature.

For nonpolar compounds, $f(T_r)$ is related to the acentric factor, ω

$$f(T_r) = [1 + m(1 - \sqrt{T_r})]^2 \quad (\text{A3})$$

$$m = 0.384401 + 1.52276\omega - 0.213808\omega^2 + 0.034616\omega^3 - 0.001976\omega^4 \quad (\text{A4})$$

For polar compounds, the Mathias–Copeman expression²⁶ is used

$$\text{for } T_r \leq 1 \quad f(T_r) = [1 + c_1(1 - \sqrt{T_r}) + c_2(1 - \sqrt{T_r})^2 + c_3(1 - \sqrt{T_r})^3]^2 \quad (\text{A5})$$

$$\text{for } T_r \geq 1 \quad f(T_r) = [1 + c_1(1 - \sqrt{T_r})]^2 \quad (\text{A6})$$

where c_1 – c_3 are pure-compound-specific parameters determined by fitting vapor pressure data.

The pure covolume parameter, b , is given by

$$b = 0.0778 \frac{RT_c}{P_c} \quad (\text{A7})$$

The translation t in eq A1, which does not, of course, affect VLE and LLE calculations, can be determined from the critical properties and the acentric factor.¹⁵ Following Pfohl,⁵⁷ however, only the temperature-independent part of the translation should be used. The translation can also be calculated as the difference between the experimental value and the value calculated from the EoS liquid molar volume at one temperature, e.g., at $T_r = 0.7$.

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